nanoRMS：同时定位不同RNA修饰

安装

wget https://github.com/novoalab/nanoRMS/archive/refs/tags/v1.0.tar.gz && tar -zxvf v1.0.tar.gz

环境配置

###docker build -f /home/chen/jiangjing/soft/nanoRMS-1.0/docker/Dockerfile . （不可用，内置有问题）

###linux配置所需依赖包

conda create nanoRMS

conda activate nanoRMS

conda install bioconda/label/cf201901::ont-tombo hdf5 minimap2 dask nanopolish

pip install "numpy<1.20" ont_fast5_api matplotlib mappy numba pysam pandas seaborn scikit-learn jupyterlab openpyxl mappy scipy

###linux R配置包

###环境：当前环境R版本低，需安装版本大于3.5，安装依赖包

①conda uninstall r-base

②conda install conda-forge::r-base

③install.packages(c("optparse","stringr","dplyr","plyr","VennDiagram","RColorBrewer","data.table"))

使用

1. prediction of RNA modified sits

1.1 Extract base-calling features using Epinano-RMS

① 对reference fasta构建索引，并且同一目录下需要有fai文件（samtools faidx）

```
java -jar epinano_RMS/picard.jar CreateSequenceDictionary
REFERENCE=reference.fasta OUTPUT= reference.fasta.dict
```

脚本示例：java -jar /home/chen/jiangjing/soft/nanoRMS/epinano_RMS/picard.jar CreateSequenceDictionary REFERENCE=yeast_all_rRNA.fa OUTPUT= reference.fasta.dict

② 计算碱基位点频率

```
python3 epinano_RMS/epinano_rms.py -R <reference_file> -b <bam_file> -s
epinano_RMS/sam2tsv
```

脚本：python3 /home/chen/jiangjing/soft/nanoRMS/epinano_RMS/epinano_rms.py -R /home/chen/jiangjing/soft/nanoRMS/predict_rna_mod/references/yeast_all_rRNA.fa -b /home/chen/jiangjing/soft/nanoRMS/predict_rna_mod/test_data/test.bam -s /home/chen/jiangjing/soft/nanoRMS/epinano_RMS/sam2tsv.jar

提醒会报错，显示无某个文件夹，直接建一个，后面删掉就行：mkdir -p ./home/chen/jiangjing/soft/nanoRMS/predict_rna_mod/test_data/test.tmp_splitted_base_freq.done_splitting

结果文件：包含每个碱基位点频率的.csv文件（test.per.site.baseFreq.csv）

```
└─[0] <> head -n20 test.per.site.baseFreq.csv
#Ref,pos,base,strand,cov,mean_q,median_q,std_q,mis,ins,del,ACGT
18s,1,T,+,2,11.500000,11.500000,4.500000,0.000000,0.000000,0.000000,0:0:0:2
18s,10,G,+,2,22.500000,22.500000,3.500000,0.000000,0.000000,0.000000,0:0:2:0
18s,100,A,+,17,12.000000,10.000000,8.538682,0.058824,0.117647,0.352941,10:0:1:0
18s,1000,C,+,37,10.675676,9.000000,6.169034,0.000000,0.189189,0.000000,0:37:0:0
18s,1001,A,+,37,13.108108,13.000000,7.043432,0.027027,0.405405,0.000000,36:0:0:1
18s,1002,G,+,37,24.756757,26.000000,8.112071,0.000000,0.243243,0.000000,0:0:37:0
18s,1003,A,+,37,24.351351,25.000000,7.121404,0.000000,0.081081,0.000000,37:0:0:0
18s,1004,T,+,37,22.189189,25.000000,7.479116,0.000000,0.054054,0.000000,0:0:0:37
```

1.2 Predict RNA modifications

① Single sample RNA modification prediction (i.e. "de novo" prediction)

##predicting pseudouridine RNA modifications in mitochondrial rRNAs, validating 2 out of the 2 sites that were predicted in all 3 biological replicates.Specifically, **pseudouridine causes strong mismatch signatures** in the modified position, largely **in the form of C-to-U mismatches**

```
Rscript --vanilla Pseudou_prediction_singlecondition.R [options] -f
<epinano_file1> (-s <epinano_file2> -t <epinano_file3>)
```

Options:

```
    -h, --help
                Show this help message and exit

    -f CHARACTER, --epinano1=CHARACTER
            first epinano file

    -s CHARACTER, --epinano2=CHARACTER
            second epinano file

    -t CHARACTER, --epinano3=CHARACTER
            third epinano file

    -p CHARACTER, --modpos=CHARACTER
            mod positions file [default=
positions/RNA_Mod_Positions_rRNAYeast.tsv]

    -m NUMBER, --misfreq=NUMBER
            Mismatch frequency threshold [default= 0.137]

    -c NUMBER, --Cfreq=NUMBER
            C mismatch frequency threshold [default= 0.578]
```

脚本示例：Rscript --vanilla /home/chen/jiangjing/soft/nanoRMS/predict_rna_mod/Pseudou_prediction_singlecondition.R -f WT_rRNA_Epinano.csv -s sn34KO_rRNA_Epinano.csv -t sn36KO_rRNA_Epinano.csv -p ../positions/RNA_Mod_Positions_rRNAYeast.tsv

##csv文件是1.1的结果文件

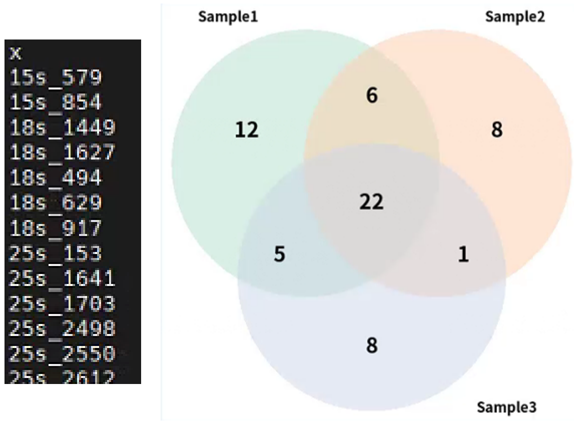
##其中mod_position文件是依据已知的假尿嘧啶修饰位点相关数据库和文献，自建的

##mod_position文件相关解释：https://github.com/novoalab/nanoRMS/issues/32

##找到的修饰位点相关数据库：https://rna.sysu.edu.cn/rmbase3/pseudosearch.php

结果文件：（1）threesample_predicted_reproducible_y_sites.tsv（三个重复样本交集位点）

（2）三个重复样本交集venn图：Threesamples_venn_diagramm_mitochondrial.tiff



① Paired sample RNA modification prediction (i.e. "differential-error"-based prediction)

##Pseudouridine is not always present in high stoichiometries (e.g. rRNAs), but can also be present in low stoichiometries (e.g. in mRNAs). Please note that pseudouridines in mRNAs cannot be accurately predicted using "de novo" mode, because the background nanopore 'error' is too similar to the 'error' caused by the presence of pseudouridine. For such cases, we can predict differentially pseudouridylated sites by identifying which sites show pseudouridine differential error signatures between two conditions, as shown below. This type of pairwise comparison can be done for WT-KO, or between two conditions (e.g. normal-heat stress). In the image below, some examples of heat-responsive sites that were identified using this script are shown. Differential-error based prediction can be applied to any type of RNA (mRNA, snoRNAs, snRNAs, rRNAs etc).

1）For Transcriptome mapped reads (Reads from one strand)

```
Rscript --vanilla Pseudou_prediction_pairedcondition_transcript.R [options] -f
<epinano_file1> -s <epinano_file2>
```

Options:

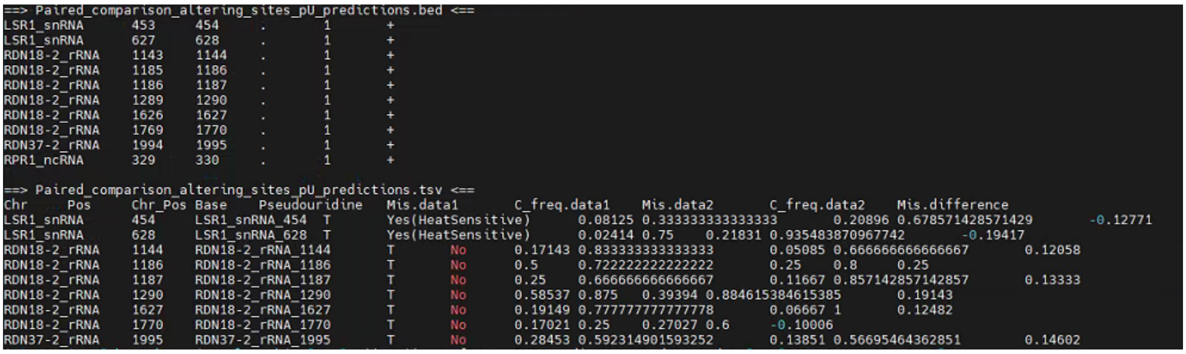
```
    -h, --help
                Show this help message and exit

    -f CHARACTER, --epinano1=CHARACTER
            first epinano file
```

```
    -s CHARACTER, --epinano2=CHARACTER
            second epinano file

    -p CHARACTER, --modpos=CHARACTER
            mod positions file
            [default=positions/RNA_Mod_Positions_ncRNAYeast_HeatSensitive.tsv]

    -m NUMBER, --misfreq=NUMBER
            Mismatch frequency threshold [default= 0.137]

    -c NUMBER, --Cfreq=NUMBER
            C mismatch frequency threshold [default= 0.578]

    -d NUMBER, --diff=NUMBER
            mismatch frequency difference threshold [default= 0.1]
```

脚本示例：Rscript --vanilla ../Pseudou_prediction_pairedcondition_transcript.R -f WT_ncRNA_Normal_Rep1_Epinano.csv -s WT_ncRNA_HeatShock_Rep1_Epinano.csv -p ../positions/RNA_Mod_Positions_ncRNAYeast_HeatSensitive.tsv

结果文件：（1）Paired_comparison_altering_sites_pU_predictions.bed

（2）Paired_comparison_altering_sites_pU_predictions.tsv



```
==> Paired_comparison_altering_sites_pU_predictions.bed <==
LSR1_snRNA      453     454     .       1       +
LSR1_snRNA      627     628     .       1       +
RDN18-2_rRNA    1143    1144    .       1       +
RDN18-2_rRNA    1185    1186    .       1       +
RDN18-2_rRNA    1186    1187    .       1       +
RDN18-2_rRNA    1289    1290    .       1       +
RDN18-2_rRNA    1626    1627    .       1       +
RDN18-2_rRNA    1769    1770    .       1       +
RDN37-2_rRNA    1994    1995    .       1       +
RPR1_ncRNA      329     330     .       1       +

==> Paired_comparison_altering_sites_pU_predictions.tsv <==
Chr     Pos     Chr_Pos Base    Pseudouridine   Mis.data1       C_freq.data1    Mis.data2       C_freq.data2    Mis.difference
LSR1_snRNA      454     LSR1_snRNA_454  T       Yes(HeatSensitive)      0.08125 0.333333333333333       0.20896 0.678571428571429       -0.12771
LSR1_snRNA      628     LSR1_snRNA_628  T       Yes(HeatSensitive)      0.02414 0.75    0.21831 0.935483870967742       -0.19417
RDN18-2_rRNA    1144    RDN18-2_rRNA_1144       T       No      0.17143 0.833333333333333       0.05085 0.666666666666667       0.12058
RDN18-2_rRNA    1186    RDN18-2_rRNA_1186       T       No      0.5     0.722222222222222       0.25    0.8     0.25
RDN18-2_rRNA    1187    RDN18-2_rRNA_1187       T       No      0.25    0.666666666666667       0.11667 0.857142857142857       0.13333
RDN18-2_rRNA    1290    RDN18-2_rRNA_1290       T       No      0.58537 0.875   0.39394 0.884615384615385       0.19143
RDN18-2_rRNA    1627    RDN18-2_rRNA_1627       T       No      0.19149 0.777777777777778       0.06667 1       0.12482
RDN18-2_rRNA    1770    RDN18-2_rRNA_1770       T       No      0.17021 0.25    0.27027 0.6     -0.10006
RDN37-2_rRNA    1995    RDN37-2_rRNA_1995       T       No      0.28453 0.592314901593252       0.13851 0.56695464362851        0.14602
```

2）For Genome mapped reads (Reads from both strands)

①csv转bed

```
./Epinano_to_BED.sh <epinano_file1>
```

脚本示例：./Epinano_to_BED.sh WT_mRNA_Normal_Epinano.csv

②Convert GTF (Only CDS) output into BED

```
Rscript --vanilla GTF_to_BED.R <GTF_File>
```

脚本示例：Rscript --vanilla ../GTF_to_BED.R Saccer64.gtf

③Intersect Epinano_BED file and GTF_BED file

```
bedtools intersect -a <Epinano_Bed> -b <GTF_Bed> -wa -wb > output.bed
```

脚本示例：bedtools intersect -a ../test_data/WT_mRNA_Normal_Epinano.csv.bed -b Saccer3.bed -wa -wb > WT_mRNA_Normal_Epinano_final.bed

④mapped

官网给的是错的：

~~Rscript --vanilla ../Pseudou_prediction_pairedcondition_genome.R -f WT_rRNA_Epinano.csv -s sn36KO_rRNA_Epinano.csv~~

第一步：修改脚本：Pseudou_prediction_pairedcondition_genome.R

```
##FUNCTION DEFINED
processing <- function(data) {
        #Import the mod file
        #mod_file <- read.delim(opt$modpos, sep="") #opt$modpos Mod positions file RNA_Mod_Positions_mRNAYeast_HeatSensitive.tsv
```

mod.file去掉#

第二步：脚本示例：

Rscript --vanilla ../Pseudou_prediction_pairedcondition_genome.R -f WT_mRNA_Normal_Epinano_final.bed -s WT_mRNA_HeatShock_Epinano_final.bed -p ../positions/RNA_Mod_Positions_mRNAYeast_HeatSensitive.tsv

结果文件：（1）Paired_comparison_altering_sites_pU_predictions.bed

（2）Paired_comparison_altering_sites_pU_predictions.tsv

```
==> Paired_comparison_altering_sites_pU_predictions.bed <==
chrI    131179  131180  .       1       +
chrI    131304  131305  .       1       +
chrI    57958   57959   .       1       -
chrI    58406   58407   .       1       -
chrII   257240  257241  .       1       -
chrII   306545  306546  .       1       -
chrII   338839  338840  .       1       -
chrII   338990  338991  .       1       -
chrII   372709  372710  .       1       -
chrII   372713  372714  .       1       -

==> Paired_comparison_altering_sites_pU_predictions.tsv <==
Chr     Pos     Chr_Pos Base    Pseudouridine   Strand.data1    cds_chr.data1   cds_start.data1 cds_end.data1   cds_strand.data1        cds_name.data1  Coverage.data1  Mis.data1       C_freq.data1    Gene_N
ame.data1       Gene_Pos.data1  Strand.data2    cds_chr.data2   cds_start.data2 cds_end.data2   cds_strand.data2        cds_name.data2  Coverage.data2  Mis.data2       C_freq.data2    Gene_Name.data2 Gene_P
os.data2        Mis.difference
chrI    131180  chrI_131180     T       No      +       chrI    130799  131980  +       YAL012W 66      0.0303  1       No      No      +       chrI    130799  131980  +       YAL012W 103     0.15534 0.9375
No      No      -0.12504
chrI    131305  chrI_131305     T       No      +       chrI    130799  131980  +       YAL012W 75      0.24    0.944444444444444       No      No      +       chrI    130799  131980  +       YAL012W 113     0
.35398  0.95    No      No      -0.11398
chrI    57959   chrI_57959      A       Yes     -       chrI    57953   58462   -       YAL044C 146     0.06164 0.888888888888889       YAL044C 504     -       chrI    57953   58462   -       YAL044C 128     0
.20312  1       YAL044C 504     -0.14148
chrI    58407   chrI_58407      A       No      -       chrI    57953   58462   -       YAL044C 99      0.20202 0.85    No      No      -       chrI    57953   58462   -       YAL044C 94      0.05319 0.8     N
o       No      0.14883
chrII   257241  chrII_257241    A       No      -       chrII   257115  257975  -       YBR011C 36      0.22222 1       No      No      -       chrII   257115  257975  -       YBR011C 107     0.11215 1       N
o       No      0.11007
chrII   306546  chrII_306546    A       No      -       chrII   306272  306955  -       YBR035C 37      0.08108 1       No      No      -       chrII   306272  306955  -       YBR035C 31      0.22581 0.8571
42857142857     No      No      -0.14473
```

~~2. RNA modification stoichiometry estimation using Nanopolish resquiggling（官网不推荐）~~

3. RNA modification stoichiometry estimation using Tombo resquiggling

3.1 第一步下载相关数据的fast5及fq文件

3.2 retrieve per-read features from all sample（fast5转bam）

```
./get_features.py --rna -f reference.fa -t 6 -i .fast5文件路径
```

结果文件：

```
batch0.fast5
batch0.fast5.bam
batch0.fast5.bam.bai
batch0.fast5.bam.json
```

3.3 Run nanopolish if you wish to compare it with tombo. You can skip this step if you don't want to compare the tools.

```
ref=reference.fa
for d in guppy3.0.3.hac/*_{snR*,wt}/; do
 if [ ! -s $d/fq.gz.bam.events.gz ]; then
  echo `date` $d;
  cat $d/*.fastq.gz > $d/fq.gz;
  nanopolish index -d $d $d/fq.gz 2> /dev/null;
  minimap2 -ax map-ont $ref $d/fq.gz 2> /dev/null | samtools sort -o
$d/fq.gz.bam;
  samtools index $d/fq.gz.bam;
  nanopolish eventalign -n -t 6 -q 10 --progress --signal-index --scale-events -
-reads $d/fq.gz --bam $d/fq.gz.bam --genome $ref | gzip >
$d/fq.gz.bam.events.gz;
 fi;
done; date
```

结果文件：

```
fq.gz
fq.gz.bam
fq.gz.bam.bai
fq.gz.bam.events.gz
```

3.4 Estimate modification frequency difference between two samples

####Note, you'll need to provide candidate positions that are likely modified. Those were identified earlier -- please see above section 1.2. Predict RNA modifications. so here we'll just generate BED file from existing candidate file.

```
# prepare BED - this is no longer needed step!
f=per_read/results/predictions_ncRNA_WT30C_WT45C.tsv.gz
zgrep -v X.Ref $f |awk -F'\t' 'BEGIN {OFS = FS} {print $1,$2-1,$2,".",100,"+"}'
> $f.bed
```

bed格式：

```
snR86_snoRNA    395    396    .    100    +
snR86_snoRNA    455    456    .    100    +
snR86_snoRNA    456    457    .    100    +
snR86_snoRNA    530    531    .    100    +
snR86_snoRNA    536    537    .    100    +
snR86_snoRNA    559    560    .    100    +
LSR1_snRNA      627    628    .    100    +
snR37_snoRNA    204    205    .    100    +
snR37_snoRNA    210    211    .    100    +
snR86_snoRNA    458    459    .    100    +
snR17a_snoRNA   211    212    .    100    +
snR37_snoRNA    115    116    .    100    +
```

```
# calculate modification frequency difference between control and sample of
interest
per_read/get_freq.py -f $ref -b $f.bed -o $f.bed.tsv.gz -1
per_read/guppy3.0.3.hac/*WT30C/workspace/*.fast5.bam -2
per_read/guppy3.0.3.hac/*WT45C/workspace/*.fast5.bam
```

usage: get_freq.py

option：

```
-h/--help
            Show this help message and exit
```

```
--version/-v

-1 control...

-2 sample...

-o output

-b BED文件

-f/--fasta

-m mincov (required minimum number of reads,but it has to be at least 5 due to KNN requirements.)
```

结果文件：predictions_ncRNA_WT30C_WT45C.bed.tsv.gz

(KNN 的最终目的是分类，而 K-means 的目的是给所有距离相近的点分配一个类别，也就是聚类)

chrom	pos	strand	min coverage	mod_freq diff kmeans	mod_freq diff knn
snR86_snoRNA	396	+	149	0.1076173518304733	0.434265364822822274
snR86_snoRNA	456	+	149	-0.06959907644745333	0.7271997965053711
snR86_snoRNA	457	+	149	0.6046921165398086	0.6493044005713503
snR86_snoRNA	531	+	149	0.08754182401627958	0.36742520594047784
snR86_snoRNA	537	+	149	0.6934275148218444	0.6954037607372767
snR86_snoRNA	560	+	149	-0.44714031345999566	0.6580507562564815
LSR1_snRNA	628	+	278	0.08354108792174181	0.3203694470166441
snR37_snoRNA	205	+	1858	0.3519216405614991	0.6108988159311087
snR37_snoRNA	211	+	1858	0.5405702862415149	0.6337119131829265
snR86_snoRNA	459	+	149	-0.6047508169135344	0.6551353043614377
snR17a_snoRNA	212	+	324	0.5894798157109572	0.5754393733329116
snR37_snoRNA	116	+	1857	0.02025265679777566	0.44149275218972883

4. Visualization of per-read current intensities at individual sites

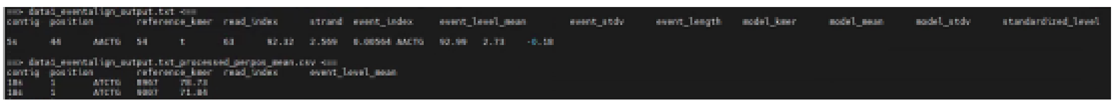
4.1 Data pre-processing

###Firstly, generate a collapsed Nanopolish event align output(3.3结果文件), by collapsing all the multiple observations for a given position from a same read.

```
python3 per_read_mean.py <event_align_file>
```

脚本示例：python3 per_read_mean.py test_data/data1_eventalign_output.txt

输出文件：data1_eventalign_output.txt_processed_perpos_mean.csv



###Secondly, create 15-mer windows of per-read current intensities centered in positions of interest

```
Rscript --vanilla nanopolish_window.R positions_file <input_table> <label>
```

脚本示例：Rscript --vanilla nanopolish_window.R test_data/positions.tsv test_data/data1_eventalign_output.txt_processed_perpos_mean.csv data1

输出文件：data1_window_file.tsv：

```
head -n5 data1_window_file.tsv
contig  position        reference_kmer  read_index      event_level_mean        Pos     sample  modification    reference
236208  25s     2819    ATAAC   39339   85.84   25s_2819        data1   25s_2826        -7
236209  25s     2819    ATAAC   39401   81.92   25s_2819        data1   25s_2826        -7
236210  25s     2819    ATAAC   39618   86.58   25s_2819        data1   25s_2826        -7
236211  25s     2819    ATAAC   39695   82.07   25s_2819        data1   25s_2826        -7
```

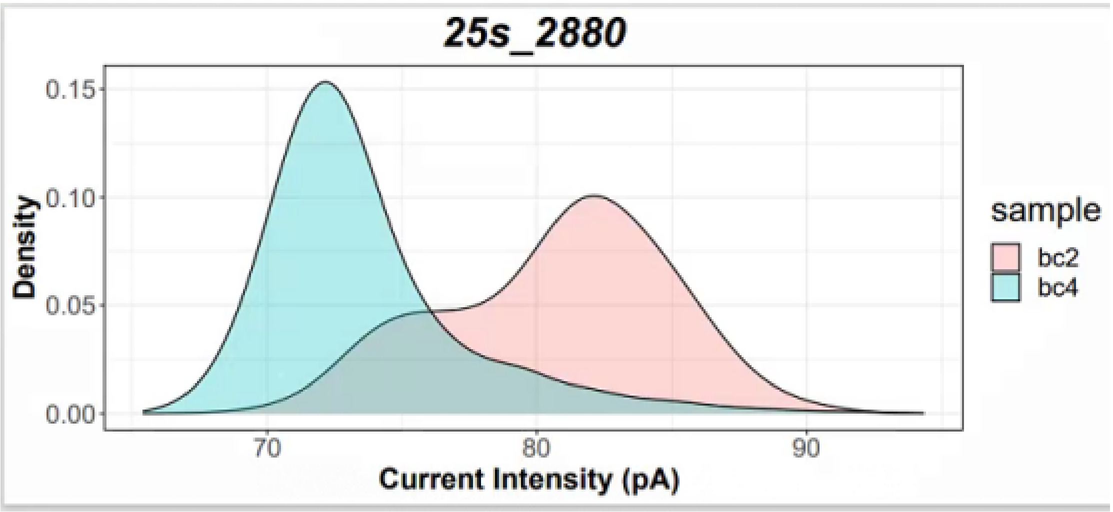
4.2. Visualization of current intensity information:

4.2.1 Density plots

```
Rscript --vanilla density_nanopolish.R <window_file1> <window_file2>
<window_file3(optional)> <window_file4(optional)>
```

脚本示例：Rscript --vanilla nanopolish_density_plot.R bc2_window_file.tsv bc4_window_file.tsv

输出文件：

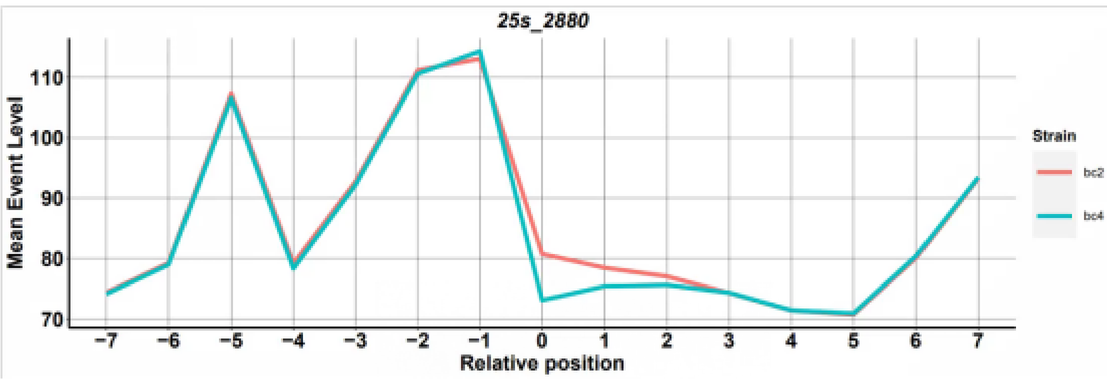


4.2.2. Mean current intensity plots centered in the modified sites

```
Rscript --vanilla nanopolish_meanlineplot.R <window_file1> <window_file2>
<window_file3(optional)> <window_file4(optional)>
```

脚本示例：Rscript --vanilla nanopolish_meanlineplot.R bc2_window_file.tsv bc4_window_file.tsv

输出文件：

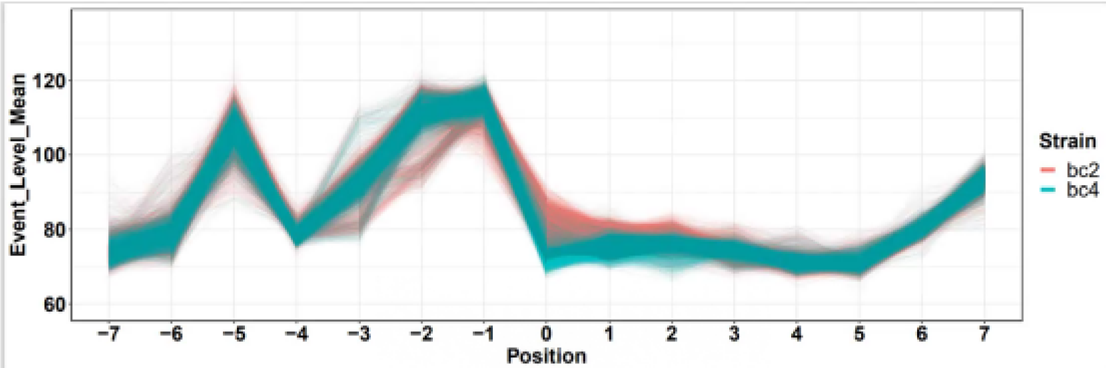


4.2.3 Per-read current intensity plots centered in the modified sites

```
Rscript --vanilla nanopolish_perreadlineplot.R <window_file1> <window_file2>
<window_file3(optional)> <window_file4(optional)>
```

脚本示例：Rscript --vanilla nanopolish_perreadlineplot.R bc2_window_file.tsv bc4_window_file.tsv

输出文件：



4.2.4 PCA plots from the per-read 15-mer current intensity data

```
Rscript --vanilla nanopolish_pca.R <window_file1.tsv> <window_file2.tsv>
<window_file3.tsv(optional)> <window_file4.tsv(optional)>
```

脚本示例：Rscript --vanilla nanopolish_pca.R bc2_window_file.tsv bc4_window_file.tsv

输出文件：